Fintech Engineering — Finance Knowledge Roadmap

4 months · 4 courses + targeted reading + 1 capstone project · Specific to bank/fintech SWE roles

- Goal: Financial domain knowledge that gets you through resume screens and interviews at banks/fintechs (risk, compliance, surveillance, capital markets engineering roles).
- What this is NOT: MBA valuation, accounting, M&A;, or behavioral finance. That's the long-term roadmap. This is the short path to job-ready.
- Your edge: 5 yrs backend at Morgan Stanley + Pinterest already satisfies the 'banking environment' requirement most candidates lack. You're filling in domain knowledge, not starting from zero.
- Weekly commitment: ~5 hrs/week alongside your Java full-stack prep and McDaniel coursework.

Phase 1 — Markets, Instruments & Institutions

Weeks 1–3 · The vocabulary and conceptual foundation every fintech/bank interview assumes

Wk 1–3 — Financial Markets

Yale University · Prof. Robert Shiller (Nobel laureate) · Coursera · 3 weeks at 10 hrs/week

Coursera ■ ~30 hrs total (do at 2x speed where possible → ~18–20 hrs effective)

- Risk, insurance, diversification, CAPM — how markets price risk
- Equities: what stocks are, dividends, market cap, how exchanges work
- Debt & bonds: what fixed income is, yield, duration, the yield curve
- Options & futures: what derivatives are conceptually, why they exist
- Banking & investment banking: how banks make money, what IBs do
- Recessions, bubbles, regulation: 2008 crisis explained clearly
- Module 5 (options/bonds) and Module 6 (IB/exchanges) are highest priority for your target roles

✓ Output: Vocabulary mastery

After this course you can speak fluently about equities, fixed income, derivatives, and market structure in an interview — the vocabulary that appears in every fintech/bank JD.

Phase 2 — How the Financial Plumbing Actually Works

Weeks 4–5 · The 'how Morgan Stanley actually works' layer — directly relevant to your resume story

Wk 4–5

Economics of Money and Banking

Columbia University · Prof. Perry G. Mehrling · Coursera · 3 weeks at 10 hrs/week (do in 2)

Coursera ■ ~20 hrs total

- How banks' balance sheets work: assets, liabilities, reserves, deposits
- The hierarchy of money: central bank → commercial banks → shadow banks
- Repos, reverse repos, money markets — the overnight plumbing of finance
- Fed funds rate, LIBOR, SOFR — what rate benchmarks are and why they matter
- Dealer banks and market-making: exactly what firms like Morgan Stanley do
- Global dollar system: why USD is the world's reserve currency operationally
- Shadow banking: why the 2008 crisis happened in money markets, not just mortgages

✓ Output: Resume narrative

This course retroactively explains what the systems you built at Morgan Stanley were actually doing. Lets you talk about your Morgan Stanley years with financial depth, not just technical depth — a major differentiator.

Phase 3

Trade Lifecycle & Compliance/Surveillance Concepts

Weeks 6–7 · Free targeted reading — maps directly to the job description you shared

Wk 6–7

Trade Lifecycle + Surveillance Reading Sprint

Investopedia · SEC/FINRA websites · Industry articles · Free

Reading ■ ~8–10 hrs total (structured reading, not passive browsing)

- Trade lifecycle: execution → clearing → settlement → custody. Know T+1 vs T+2.
- Prime brokerage: what it is, why hedge funds use it, Morgan Stanley's role
- Market misconduct types: spoofing, layering, front-running, wash trading, marking the close
- What trade surveillance systems detect and why regulators mandate them
- Key regulators: SEC, FINRA, CFTC — their mandates and what they look for
- The Quartz platform (Goldman internal) — search 'Goldman Sachs Quartz platform' and read publicly available tech articles about it
- MiFID II and Dodd-Frank: the two major post-2008 regulations that drove surveillance tech spending

✓ Output: Interview talking points

The job posting you shared (Enterprise Risk & Finance Technology, Quartz platform, surveillance/compliance) maps exactly to this reading list. Knowing what spoofing is, how T+1 settlement works, and what Dodd-Frank requires is what separates candidates in domain interviews.

Phase 4

Derivatives, Fixed Income & Risk Measurement

Weeks 8–14 · Graduate-level rigor — the quantitative finance behind the systems you'd build

Financial Engineering and Risk Management Specialization

Wk 8–14

Columbia University · Profs. Martin Haugh & Garud Iyengar · 5 courses · Coursera · ~18 weeks at self-pace (compress to 7 weeks at 8 hrs/week)

Coursera ■ ~55–60 hrs total across 5 courses

- Course 1 — Fixed income basics, bond pricing, term structure, swaps, options intro, Black-Scholes
- Course 2 — Equity derivatives: Black-Scholes deep dive, Greeks (Delta/Gamma/Vega), delta hedging
- Course 3 — Credit derivatives: CDOs, credit default swaps (CDS), structured products — exactly what broke in 2008
- Course 4 — Portfolio optimization: mean-variance, factor models, asset allocation
- Course 5 — Algorithmic trading & risk: backtesting trading signals, risk measurement, VaR
- Prerequisites honestly: you need comfort with probability and some calculus. Your CS background covers it.
- Don't aim for perfect math mastery — aim to understand WHAT these instruments are and WHY firms use them

✓ Output: Technical depth certificate

This is the course that signals to hiring managers at Goldman, Morgan Stanley, JPMorgan, and fintechs that you understand the domain at an engineering level — not just as a user. The Columbia brand on the certificate matters in this specific industry.

Phase 5 Capstone Project — Visible, GitHub-Ready Output

Weeks 15–17 · This is what turns knowledge into a job offer

Build: Trade Surveillance Signal Detector (Java or Python)

Wk 15–17

Personal project · GitHub · ~15–20 hrs

Build ■ ~15–20 hrs (build it in whichever language your target roles require)

- Ingest a public market dataset (e.g. from Yahoo Finance API or SEC EDGAR trade data)
- Implement 2–3 detection rules: spoofing (large orders cancelled before execution), wash trading (same account buys and sells), unusual volume spikes near close
- Build a simple REST API that accepts trade data and returns flagged events — mirrors exactly what surveillance platforms do
- Add a basic dashboard (even a terminal output) showing flagged trades with explanations
- Write a README that explains the financial concepts behind each detection rule
- Use Java if targeting Java roles, Python if targeting Python roles — or build both with a shared core

✓ Output: Portfolio project

This project directly mirrors what the job posting you shared asks for. When an interviewer asks 'do you understand what our systems do?' you show them this. No other candidate applying from a generic backend background has this. It's the single most leveraged thing you can build.

4-Month Timeline Summary

Phase	Weeks	Course / Activity	Platform	Hrs	Job-Relevant Output
1	1–3	Financial Markets — Shiller (Yale)	Coursera	~30	Market vocabulary, instruments, banking concepts
2	4–5	Economics of Money & Banking — Mehrling (Columbia)	Coursera	~20	How bank plumbing works; reframes MS experience

3	6–7	Trade lifecycle + Surveillance reading sprint	Free	~10	Compliance/surveillance interview talking points
4	8–14	Financial Engineering & Risk Mgmt Specialization (Columbia, 5 courses)	Coursera	~60	Derivatives, risk, fixed income — Columbia cert
5	15–17	Trade Surveillance Detector project (Java/Python + REST API)	Build	~20	GitHub portfolio piece mirroring target roles
TOTAL	**~17 weeks**	**4 courses + 1 specialization + 1 project**	**—**	**~140 hrs**	**Job-ready for bank/fintech SWE roles**

After this roadmap, continue with:

• The MBA Finance roadmap (long-term, 18–24 months) — now it enriches your career rather than blocking it

• FINRA Series 99 (Operations Professional exam) — optional but signals serious commitment for capital markets roles

• CFA Level 1 — if you later want to move toward quant or product roles at fintechs; pairs well with your McDaniel AI track